Leonid Petrov. Publication List

Preprints (∗), accepted, or published works

[54] (∗) Leonid Petrov. *From Yang-Baxter to Robinson-Schensted-Knuth*, arXiv:2609.18502 [math.CO]. (PDF link). 58 pages.

[53] (∗) Leonid Petrov, Jiaming Xu. *Perturbed Beta Corners Process*, arXiv:2607.27810 [math.PR]. (PDF link). 31 pages.

[52] (∗) Leonid Petrov. *A Borodin–Okounkov–Geronimo–Case identity for tilted Toeplitz minors*, arXiv:2605.24976 [math.FA]. (PDF link). 34 pages.

[51] David Anderson, Greta Panova, Leonid Petrov. *Computation and sampling for Schubert specializations*, **Proc. Natl. Acad. Sci. USA 123 (2026), no. 44, e2615757123**. arXiv:2603.20104 [math.CO]. (PDF link). 33 pages.

[50] Natasha Blitvić, Leonid Petrov. *Colored Interlacing Triangles and Genocchi Medians*, **SIGMA 22 (2026), 081, 30 pages**. arXiv:2602.04390 [math.CO]. (PDF link). 30 pages.

[49] (∗) Alisa Knizel, Leonid Petrov. *Random Lozenge Waterfall: Dimensional Collapse of Gibbs Measures*, arXiv:2507.22011 [math.PR]. (PDF link). 65 pages.

[48] (∗) Alexey Bufetov, Leonid Petrov, Panagiotis Zografos. *Domino Tilings of the Aztec Diamond in Random Environment and Schur Generating Functions*, arXiv:2507.08560 [math.PR]. (PDF link). 33 pages.

[47] Leonid Petrov, Jeanne Scott. *Random Fibonacci Words via Clone Schur Functions*, **Forum Math. Sigma 14 (2026), e15**. arXiv:2412.21126 [math.PR]. (PDF link). 92 pages.

[46] Greta Panova, Leonid Petrov. *Hook-length Formulas for Skew Shapes via Contour Integrals and Vertex Models*, **Annales de l'Institut Henri Poincaré D: Combinatorics, Physics and their Interactions (2026), published online first**. arXiv:2409.17842 [math.CO]. (PDF link). 33 pages.

[45] Alejandro H. Morales, Greta Panova, Leonid Petrov, Damir Yeliussizov. *Grothendieck Shenanigans: Permutons from Pipe Dreams via Integrable Probability*, **Advances in Mathematics, 480 (2025), Part C, 110510.**. **FPSAC 2025 poster**. arXiv:2407.21653 [math.PR]. (PDF link). 50 pages.

[44] Amol Aggarwal, Matthew Nicoletti, Leonid Petrov. *Colored Interacting Particle Systems on the Ring: Stationary Measures from Yang–Baxter Equation*, **Compositio Math. 161 (2025), no. 8, 1855–1922**. arXiv:2309.11865 [math.PR]. (PDF link). 73 pages.

[43] Svetlana Gavrilova, Leonid Petrov. *Tilted biorthogonal ensembles, Grothendieck random partitions, and determinantal tests*, **Selecta Math. (2024), Volume 30, article 56**. arXiv:2305.17747 [math.PR]. (PDF link). 47 pages.

[42] Leonid Petrov, Mikhail Tikhonov. *Asymptotics of noncolliding q-exchangeable random walks*, **J. Phys. A: Math. Theor. 56 365203**. arXiv:2303.02380 [math.PR]. (PDF link). 34 pages.

[41] Leonid Petrov, Axel Saenz. *Rewriting History in Integrable Stochastic Particle Systems*, **Commun. Math. Phys., 405 (300), 2024**. arXiv:2212.01643 [math.PR]. (PDF link). 76 pages.

[40] Leonid Petrov. *Noncolliding Macdonald walks with an absorbing wall*, **SIGMA 18 (2022), 079, 21 pages**. arXiv:2204.09206 [math.PR]. (PDF link). 21 pages.

[39] Matthew Nicoletti, Leonid Petrov. *Irreversible Markov Dynamics and Hydrodynamics for KPZ States in the Stochastic Six Vertex Model*, **Electronic Journal of Probability 2023, Vol. 28, paper no. 138, 1-55.**. arXiv:2201.12497 [math.PR]. (PDF link). 59 pages.

[38] Amol Aggarwal, Alexei Borodin, Leonid Petrov, Michael Wheeler. *Free Fermion Six Vertex Model: Symmetric Functions and Random Domino Tilings*, **Selecta Math., 29, article 36 (2023)**. arXiv:2109.06718 [math.PR]. (PDF link). 138 pages.

[37] Leonid Petrov. *Refined Cauchy identity for spin Hall-Littlewood symmetric rational functions*, **Journal of Combinatorial Theory Ser. A, vol. 184 (2021), 105519**. arXiv:2007.10886 [math.CO]. (PDF link). 40 pages.

[36] Matteo Mucciconi, Leonid Petrov. *Spin q-Whittaker polynomials and deformed quantum Toda*, **Communications in Mathematical Physics, 389, pages 1331-1416 (2022)**. arXiv:2003.14260 [math.PR]. (PDF link). 86 pages.

[35] Leonid Petrov, Mikhail Tikhonov. *Parameter symmetry in perturbed GUE corners process and reflected drifted Brownian motions*, **Journal of Statistical Physics, 181 (2020), 1996-2010**. arXiv:1912.08671 [math.PR]. (PDF link). 15 pages.

[34] Leonid Petrov. *Parameter permutation symmetry in particle systems and random polymers*, **SIGMA 17 (2021), 021, 34 pages**. arXiv:1912.06067 [math.PR]. (PDF link). 34 pages.

[33] Leonid Petrov. *PushTASEP in inhomogeneous space*, **Electronic Journal of Probability, vol. 25 (2020), paper no. 114**. arXiv:1910.08994 [math.PR]. (PDF link). 26 pages.

[32] Leonid Petrov, Axel Saenz. *Mapping TASEP back in time*, **Probability Theory and Related Fields, 182, pages 481-530 (2022)**. arXiv:1907.09155 [math.PR]. (PDF link). 50 pages.

[31] Alexey Bufetov, Matteo Mucciconi, Leonid Petrov. *Yang-Baxter random fields and stochastic vertex models*, **Advances in Mathematics 388 (2021), 107865**. arXiv:1905.06815 [math.PR]. (PDF link). 94 pages.

[30] Ivan Corwin, Konstantin Matveev, Leonid Petrov. *The q-Hahn PushTASEP*, **International Mathematics Research Notices (2019), rnz106**. arXiv:1811.06475 [math.PR]. (PDF link). 40 pages.

[29] Alisa Knizel, Leonid Petrov, Axel Saenz. *Generalizations of TASEP in discrete and continuous inhomogeneous space*, **Communications in Mathematical Physics 372 (2019), no. 3, pp 797-864**. arXiv:1808.09855 [math.PR]. (PDF link). 68 pages.

[28] Christian Gromoll, Mark Meckes, Leonid Petrov. *Quenched Central Limit Theorem in a Corner Growth Setting*, **Electronic Communications in Probability (2018), Vol. 23, paper no. 101, 1-12**. arXiv:1804.04222 [math.PR]. (PDF link). 12 pages.

[27] Alexey Bufetov, Leonid Petrov. *Yang-Baxter field for spin Hall-Littlewood symmetric functions*, **Forum of Mathematics Sigma 7 (2019), e39**. arXiv:1712.04584 [math.PR]. (PDF link). 70 pages.

[26] Michael Damron, Leonid Petrov, David Sivakoff. *Coarsening model on Zd with biased zero-energy flips and an exponential large deviation bound for ASEP*, **Communications in Mathematical Physics 362 (2018), no. 1, 185-217**. arXiv:1708.05806 [math.PR]. (PDF link). 33 pages.

[25] Sevak Mkrtchyan, Leonid Petrov. *GUE corners limit of q-distributed lozenge tilings*, **Electronic Journal of Probability, Volume 22 (2017), paper no. 101, 24 pp**. arXiv:1703.07503 [math.PR]. (PDF link). 24 pages.

[24] Alexei Borodin, Leonid Petrov. *Inhomogeneous exponential jump model*, **Probability Theory and Related Fields 172 (2018), 323-385**. arXiv:1703.03857 [math.PR]. (PDF link). 63 pages.

[23] Daniel Orr, Leonid Petrov. *Stochastic higher spin six vertex model and q-TASEPs*, **Advances in Mathematics 317 (2017), 473-525**. arXiv:1610.10080 [math.PR]. (PDF link). 53 pages.

[22] Vadim Gorin, Leonid Petrov. *Universality of local statistics for noncolliding random walks*, **Annals of Probability (2019), Vol. 47, No. 5, 2686-2753**. arXiv:1608.03243 [math.PR]. (PDF link). 68 pages.

[21] Alexei Borodin, Leonid Petrov. *Lectures on Integrable probability: Stochastic vertex models and symmetric functions*, **Lecture Notes of the Les Houches Summer School, Volume 104, July 2015**. arXiv:1605.01349 [math.PR]. (PDF link). 113 pages.

[20] Alexei Borodin, Leonid Petrov. *Higher spin six vertex model and symmetric rational functions*, **Selecta Mathematica 24 (2018), no. 2, 751-874**. arXiv:1601.05770 [math.PR]. (PDF link). 124 pages.

[19] Konstantin Matveev, Leonid Petrov. *q-randomized Robinson-Schensted-Knuth correspondences and random polymers*, **Annales de l'Institut Henri Poincare D: Combinatorics, Physics and their Interactions 4 (2017), no. 1, 1-123**. arXiv:1504.00666 [math.PR]. (PDF link). 123 pages.

[18] Ivan Corwin, Leonid Petrov. *Stochastic higher spin vertex models on the line*, **Communications in Mathematical Physics 343 (2016), no. 2, 651-700**. arXiv:1502.07374 [math.PR]. (PDF link). 50 pages. [Includes correction: **Commun. Math. Phys. 371, 353–355 (2019)**.]

[17] Alexei Borodin, Ivan Corwin, Leonid Petrov, Tomohiro Sasamoto. *Spectral theory for interacting particle systems solvable by coordinate Bethe ansatz*, **Communications in Mathematical Physics 339 (2015), no. 3, 1167-1245**. arXiv:1407.8534 [math-ph]. (PDF link). 79 pages. [Includes correction: **Commun. Math. Phys. 370, 1069–1072 (2019)**.]

[16] Alexey Bufetov, Leonid Petrov. *Law of Large Numbers for Infinite Random Matrices over a Finite Field*, **Selecta Mathematica 21 (2015), no. 4, 1271-1338**. arXiv:1402.1772 [math.PR]. (PDF link). 68 pages.

[15] Alexei Borodin, Leonid Petrov. *Integrable probability: From representation theory to Macdonald processes*, **Probability Surveys, 11 (2014), 1-58**. arXiv:1310.8007 [math-ph]. (PDF link). 58 pages.

[14] Alexei Borodin, Ivan Corwin, Leonid Petrov, Tomohiro Sasamoto. *Spectral theory for the q-Boson particle system*, **Compositio Mathematica, 151 (2015), no. 1, 1-67**. arXiv:1308.3475 [math-ph]. (PDF link). 67 pages.

[13] Ivan Corwin, Leonid Petrov. *The q-PushASEP: A New Integrable Model for Traffic in 1+1 Dimension*, **Journal of Statistical Physics, 160 (2015), no. 4, 1005-1026**. arXiv:1308.3124 [math.PR]. (PDF link). 22 pages.

[12] Alexei Borodin, Leonid Petrov. *Nearest neighbor Markov dynamics on Macdonald processes*, **Advances in Mathematics, 300 (2016), 71-155**. arXiv:1305.5501 [math.PR]. (PDF link). 85 pages.

[11] Leonid Petrov. *The Boundary of the Gelfand-Tsetlin Graph: New Proof of Borodin-Olshanski's Formula, and its q-analogue*, **Moscow Mathematical Journal, 14 (2014) no. 1, 121-160**. arXiv:1208.3443 [math.CO]. (PDF link). 36 pages.

[10] Leonid Petrov. *Asymptotics of uniformly random lozenge tilings of polygons. Gaussian free field*, **Annals of Probability, 43 (2014), no. 1, 1-43**. arXiv:1206.5123 [math.PR]. (PDF link). 43 pages.

[9] Leonid Petrov. *Asymptotics of Random Lozenge Tilings via Gelfand-Tsetlin Schemes*, **Probability Theory and Related Fields, 160 (2014), no. 3, 429-487**. arXiv:1202.3901 [math.PR]. (PDF link). 59 pages.

[8] Leonid Petrov. *sl(2) Operators and Markov Processes on Branching Graphs*, **Journal of Algebraic Combinatorics 38 (2013), no. 3, 663-720**. arXiv:1111.3399 [math.CO]. (PDF link). 58 pages.

[7] Leonid Petrov. *On Measures on Partitions Arising in Harmonic Analysis for Linear and Projective Characters of the Infinite Symmetric Group*, **Proceedings of the international conference "50 years of IITP"**. arXiv:1107.0676 [math.CO]. (PDF link). 7 pages.

[6] Leonid Petrov. *Pfaffian Stochastic Dynamics of Strict Partitions*, **Electronic Journal of Probability 16 (2011), 2246-2295**. arXiv:1011.3329 [math.PR]. (PDF link). 50 pages.

[5] Leonid Petrov. *Random Strict Partitions and Determinantal Point Processes*, **Electronic Communications in Probability 15 (2010), 162-175**. arXiv:1002.2714 [math.PR]. (PDF link). 14 pages.

[4] Leonid Petrov. *Random Walks on Strict Partitions*, **Journal of Mathematical Sciences 168 (2010), no. 3, 437-463**. arXiv:0904.1823 [math.PR]. (PDF link). 27 pages.

[3] Leonid Petrov. *Limit Behavior of Certain Random Walks on Strict Partitions*, **Russian Mathematical Surveys 64 (2009), no. 6, 1139-1141**. (PDF link). 3 pages.

[2] Leonid Petrov. *A Two-parameter Family of Infinite-dimensional Diffusions in the Kingman Simplex*, **Functional Analysis and Its Applications 43 (2009), no. 4, 279-296**. arXiv:0708.1930 [math.PR]. (PDF link). 18 pages.

[1] Leonid Petrov. *Asymptotic Behavior of a Certain Collection of Particles on a Line Under Synchronization*, **Proceedings of the XXVIII Conference of Young Scientists of Department of Mechanics and Mathematics of the Lomonosov Moscow State University (2006), 152-156, in Russian**. (PDF link). 4 pages.

Lecture Notes & Expository Works

[7] Random matrices, graduate topics course at University of Virginia, Spring 2025. Course webpage with lecture notes. (PDF link), 3.4MB, 232 pages.

[6] Asymptotic Representation Theory, graduate topics course at University of Virginia, Fall 2022. Course webpage with lecture notes. (PDF link), 102 MB.

[5] Particle systems, graduate topics course at University of Virginia, Spring 2021. Lecture notes in Obsidian format on github. (PDF link), 312 MB.

[4] Random matrices, graduate topics course at University of Virginia, Fall 2019. Course webpage with lecture notes. (PDF link), 38 MB.

[3] Random matrices, graduate topics course at University of Virginia, Spring 2016. Lecture notes on github. (PDF link), 700 KB.

[2] Topics in Probability, graduate topics course at Northeastern University, Fall 2012. Lecture notes at AMS Open Notes. (PDF link), 1.7 MB.

[1] Course notes "Asymptotic representation theory", taught by G. Olshanski at the Independent University of Moscow, Fall 2009 and Spring 2010 (in Russian) . Webpage with notes. (PDF link), 2.6 MB.

Other works

[1] Sihan Li, Andrew Mecca, Jeewoo Kim, Giusy Caprara, Elizabeth Wagner, Ting-Ting Du, Leonid Petrov, Wenhao Xu, Runjia Cui, Ivan Rebustini, Bechara Kachar, Anthony Peng, and Jung-Bum Shin, *Myosin-VIIa is expressed in multiple isoforms and essential for tensioning the hair cell mechanotransduction complex*. **Nature Communications, 11, Article number: 2066 (2020).** (PDF link). 15 pages. (Contribution: developed a mathematical model for reconstructing stereocilia length measurements from electron microscopy observations.)